Lattice

Asset Tracker & Budget Intelligence Platform

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Status: Draft — Internal Review

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Audience: Engineering, Design

Platform: Web App (Browser-native) + Local AI Layer

1. Product Overview

Lattice is a private, household-scale asset tracker and budget intelligence platform. It consolidates net worth across cash, bank accounts, crypto (via xpub), and stocks into a single real-time dashboard. An embedded AI avatar — blunt, data-driven, no fluff — surfaces insights on spending patterns, allocation drift, and projected financial health across a configurable rolling window.

Lattice is built for personal use with shared household access. It is not a fintech product targeting third-party users. All data is owner-controlled.

1.1 Problem Statement

Managing multiple asset classes across wallets, banks, and brokerage accounts produces a fragmented view of actual net worth. Budget planning is done manually or not at all. No single tool combines real-time crypto tracking (xpub-derived), live stock pricing, manual fiat entries, and forward-looking budget projections — especially not with a local AI layer that interprets the data without sugarcoating.

1.2 Vision

One dashboard. Every asset. Honest projections. An AI analyst that tells you exactly where you stand and where you are headed if you keep spending the way you do.

2. Users & Personas

Persona	Role	Key Needs
Primary Owner	Full admin access	Full dashboard, AI insights, budget configuration, all asset management, CSV import, xpub setup
Household Member	Read + limited write	View net worth, log manual transactions, see spend limits — no budget or asset config access

Multi-user access is scoped by role. The primary owner configures the system; household members consume it. No third-party accounts. No social features.

3. Core Features

3.1 Asset Management

Cash / Wallet

- Manually enter cash holdings in any currency
- Label by wallet name (e.g., 'Physical Wallet PH', 'USD Envelope')
- Update balance at any time; history logged with timestamp

Bank Accounts

- Add accounts manually with name, bank, currency, and current balance
- Update balance manually or via CSV import
- CSV import: map columns (date, description, amount, balance) on first import; mapping saved per account
- Running transaction log per account

Crypto — BTC & LTC via xpub

- Enter xpub key; system derives addresses and fetches balance via public blockchain APIs
- Balance auto-refreshes on a configurable interval (default: every 15 minutes)
- Displays: current balance in BTC/LTC, fiat equivalent, 24h change
- No private keys stored or transmitted — xpub is read-only by design
- Supports BIP44/BIP49/BIP84 derivation paths

Stocks

- Search and select any ticker (global markets via live price API)
- Enter number of shares held
- Live price polling updates fiat value in real time (configurable interval; default 5 min during market hours)
- Displays: current price, shares, total value, daily P&L, % change
- Multiple positions per ticker supported

3.2 Multi-Currency Support

- All assets stored in native currency; converted to a user-selected base display currency
- Exchange rates fetched from a public FX API (e.g., Frankfurter, Open Exchange Rates)
- Base currency switchable at any time — all values recompute instantly
- Supported currencies: PHP, USD, EUR, SGD, JPY, GBP, AUD, and any ISO 4217 currency available from the FX provider
- Crypto fiat conversion uses CoinGecko or equivalent public API

3.3 Net Worth Dashboard

- Total net worth displayed prominently at top, broken down by asset class
- Asset allocation donut chart — cash, bank, crypto, stocks
- Historical net worth chart — configurable range (7d, 30d, 90d, 1y, custom)
- Drill-down per asset class: click into any category to see line-item breakdown
- Last-updated timestamps per data source

3.4 Budget Planning & Projections

Income Setup

- Define income sources: name, amount, currency, frequency (weekly, biweekly, monthly, irregular)
- Irregular income: enter actuals as they arrive; system uses rolling average for projections
- Multiple income streams supported

Spend Category Configuration

- Create custom categories (e.g., Food, Transport, Training, Subscriptions, Utilities)
- Set monthly spend limits per category
- Assign transactions to categories manually or via auto-tagging rules (keyword-based)

Custom Rolling Window Projections

- User defines projection window: 1 month minimum, no hard maximum
- Projection engine computes: expected income total, projected spend at current burn rate, projected net savings, projected net worth at end of window
- Burn rate derived from trailing N-week average (user-configurable, default: 4 weeks)
- Scenario modeling: adjust projected income or spend to see outcome changes
- Projections update live as transactions are logged

Net Income Flow

- Monthly summary: total income in, total spend out, net
- Cash flow chart: income vs. spend per month, last 12 months
- Category breakdown: spend per category vs. limit, color-coded (under/at/over)

3.5 AI Avatar (Name: Tess) — Analyst Mode

Behavior

- Personality: blunt analyst — states facts, flags problems, no encouragement theater

- Runs locally in-browser via the Web AI API (window.ai / Chrome Built-in AI) where available; falls back to Anthropic API call if local inference unavailable
- Avatar rendered as a minimal visual presence in the dashboard — not a chatbot widget, more of a persistent commentary layer

Insight Types

- Net worth commentary: 'Your net worth dropped 4.2% this week. Stocks account for 80% of that.'
- Spend limit breaches: immediate flag when a category exceeds its limit
- Projection warnings: 'At your current burn rate, you will overshoot your budget by [X] by end of window.'
- Allocation drift: 'Crypto is now 61% of your portLattice. That is above your prior 40% target.'
- Inactivity nudges: 'No transactions logged in 8 days. Data is stale.'
- Weekly digest: auto-generated summary of the past 7 days — net worth delta, top spend categories, projection status

User Interaction

- Ask the avatar a direct question (text input): 'How much can I spend on food this month and still hit my savings target?'
- Avatar has read access to all dashboard state — net worth, transactions, budget config, projections
- No write access — avatar cannot modify data
- Conversation history persisted per session; no cross-session memory by default

4. Information Architecture

The app is structured as a multi-section layout with drill-down capability. Navigation is persistent sidebar on desktop.

Section	Route	Description
Dashboard	/	Net worth summary, allocation chart, AI avatar digest, quick stats
Assets	/assets	All asset classes — cash, bank, crypto, stocks — with drill-down per type
Budget	/budget	Income setup, spend categories, limits, net income flow chart
Projections	/projections	Rolling window config, projection outputs, scenario modeling
Transactions	/transactions	Unified ledger — all logged transactions, filterable by account/category/date
Settings	/settings	User accounts, currency config, API keys, xpub management, import mappings
AI Insights	/insights	Full AI avatar chat interface + history + weekly digest archive

5. Data Sources & External APIs

Data Type	Source / Approach
Stock prices	Yahoo Finance (unofficial) or Polygon.io — live polling, 5-min default during market hours
Crypto balances (BTC/LTC)	BlockCypher or Blockstream API — xpub address derivation + UTXO balance fetch
Crypto fiat price	CoinGecko public API — BTC/LTC to any fiat
FX rates	Frankfurter.app (free, ECB-backed) or Open Exchange Rates
AI inference (primary)	Chrome Built-in AI (window.ai) — local, zero latency, no API cost
AI inference (fallback)	Anthropic Claude API — triggered if window.ai unavailable
CSV import	Client-side parsing — no data leaves the browser during import

All external API calls are made from the client or a lightweight backend proxy. No financial data is sent to third-party AI services without explicit user opt-in (local AI is the default).

6. Non-Functional Requirements

Requirement	Spec
Performance	Dashboard initial load < 2s. Asset refresh (crypto/stocks) non-blocking — updates without full reload.
Data privacy	All sensitive data (xpub, balances, transactions) stored locally or in a self-hosted/owner-controlled backend. No third-party analytics by default.
Security	xpub keys stored encrypted at rest. No private keys ever accepted. Session auth required for all data access.
Offline capability	Last-known state cached locally. App remains readable offline; data sync resumes on reconnect.
Multi-currency accuracy	FX rates refreshed every 30 minutes minimum. Stale rate indicator shown if last refresh > 2 hours.
AI availability	Graceful degradation if both window.ai and Anthropic API are unavailable — avatar shows 'Offline' state, rest of app unaffected.
Household access	Role-based access enforced server-side. Household members cannot access budget config, xpub settings, or API key management.
Audit log	All balance updates, CSV imports, and budget changes logged with timestamp and user.

7. Out of Scope (v1.0)

- Open banking / Plaid / automatic bank sync (manual + CSV only at launch)
- Tax reporting or capital gains calculation
- DeFi / NFT / altcoin tracking beyond BTC and LTC
- Mobile native app (web-first; responsive design for mobile browser access)
- Social or sharing features
- Automated trades or order execution
- Third-party integrations beyond APIs listed in Section 5

8. Suggested Build Milestones

Phase	Scope	Deliverable
M1 — Foundation	Project scaffold, auth, DB schema, settings page, multi-currency config	Running shell with login + currency setup
M2 — Assets	Cash/wallet, bank accounts (manual + CSV), stocks (live price)	Asset dashboard with net worth total
M3 — Crypto	xpub integration (BTC/LTC), balance derivation, fiat conversion	Crypto asset cards in dashboard
M4 — Budget	Income setup, spend categories, limits, transaction ledger, net income chart	Budget section fully functional
M5 — Projections	Rolling window engine, scenario modeling, burn rate calc	Projections page live
M6 — AI Avatar	window.ai integration, fallback API, insight types, chat interface, weekly digest	AI avatar live in dashboard
M7 — Polish	Drill-down UX, mobile responsive, audit log, performance tuning	v1.0 shippable

9. Open Questions

Question	Decision Needed By
Backend: self-hosted (SQLite/Postgres) vs. Supabase?	M1 kickoff
Frontend framework: Next.js (existing stack) vs. pure React SPA?	M1 kickoff
Anthropic API key: owner-provided in settings vs. bundled?	M6 start
Stock API: Polygon.io (free tier limits) vs. Yahoo Finance scrape vs. paid?	M2 start
xpub storage: local browser storage vs. encrypted server-side?	M3 start
Household member invite flow: email link vs. manual credential setup?	M1 kickoff
App name: 'Lattice' confirmed or placeholder?	Pre-M1

End of Document — Lattice PRD v1.0